MTH1001-B

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2022

Mathematical Structures

Module Leader: Dr Gihan Marasingha

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No word count specified

SECTION A

1. (a) (i) Using the extended Euclidean Algorithm, find integers m and n for which $\gcd(321, 71) = 321m + 71n$.

   (ii) Thereby, find an integer x in the range $0 \leq x < 71$ for which $321x \equiv 5 \pmod{71}$. (12)

   (b) Let $\sigma = (56)(43)(16)$ and $\tau = (124)(173)$ be permutations in the symmetric group S_7.

   (i) Write each of σ and τ as products of disjoint cycles.

   (ii) Find $\sigma\tau$, $\tau\sigma$, and $\tau\sigma\tau^{-1}$, writing each as a product of disjoint cycles. Find the order and the sign of each of these permutations. (12)

   (c) Consider the following vectors in $\mathbb{R}^4$:

$$\mathbf{w}_1 \coloneqq \begin{bmatrix} 1 \\ 0 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{w}_2 \coloneqq \begin{bmatrix} 1 \\ 1 \\ 3 \\ 5 \end{bmatrix}, \quad \mathbf{w}_3 \coloneqq \begin{bmatrix} 3 \\ 1 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{w}_4 \coloneqq \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{b} \coloneqq \begin{bmatrix} 2 \\ -2 \\ -3 \\ -3 \end{bmatrix}.$$

   (i) Find a maximal linearly independent sublist, S of the list $[\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4]$. Justify your answer.

   (ii) Express $\mathbf{b}$ as a linear combination of this linearly independent list S. Justify your answer. (13)

   (d) Let V be a vector space over a field K. Suppose $\mathbf{v}_1, \mathbf{v}_2 \in V$. Prove that the list $[\mathbf{v}_1, \mathbf{v}_2]$ is linearly independent if and only if the list $[\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2]$ is linearly independent. (13)

[**50**]

SECTION B

2. (a) Let a, b, and n be integers. Give the definitions of 'a divides b' and '$a \equiv b \pmod{n}$'. (3)

   (b) For each statement below, prove either the statement or its negation. The variables a, b, c, and n are integers. The notation $x \mid y$ means 'x divides y'.

   (i) If $a \mid b$, then $a \mid bc$.

   (ii) Suppose $c \not\equiv 0 \pmod{n}$. If $ca \equiv cb \pmod{n}$, then $a \equiv b \pmod{n}$.

   (iii) If $n \mid a$ and $n \mid b$, then $n \mid a + b$. (11)

   (c) Let $(G, *)$ and $(H, \cdot)$ be groups.

   (i) What does it mean for ϕ to be a group homomorphism from G to H?

   (ii) Suppose $\phi : G \to H$ is an invertible group homomorphism. Prove that $\phi^{-1} : H \to G$ is a group homomorphism. (11)

   [25]

3. (a) (i) State the fundamental theorem of arithmetic.

   (ii) Let $n \geq 2$ be a natural number. Prove that if $\sqrt{n}$ is rational, then n is a square. (12)

   (b) Let U and V be real vector spaces. What does it mean for T to be a linear map from U to V? (2)

   (c) Prove or disprove the following statements. Here, T is a linear map between real vector spaces U and V.

   (i) If the list $[\mathbf{u}_1, \ldots, \mathbf{u}_n]$ of vectors of U is linearly independent, then $[T(\mathbf{u}_1), \ldots, T(\mathbf{u}_n)]$ is linearly independent.

   (ii) If the list $[\mathbf{u}_1, \ldots, \mathbf{u}_n]$ of vectors of U is linearly dependent, then $[T(\mathbf{u}_1), \ldots, T(\mathbf{u}_n)]$ is linearly dependent. (11)

   [25]

4. (a) (i) Let V be a vector space over a field K. Let U be a subset of V. What does it mean for U to be a *subspace* of V?

(ii) Which of the following are subspaces of $P(\mathbb{R})$, the vector space of polynomials with real coefficients? Briefly justify your answers.

$$
\begin{aligned}
S &\coloneqq \{f \in P(\mathbb{R}) \mid \deg(f) = 3\},\\
T &\coloneqq \{f \in P(\mathbb{R}) \mid \textstyle\int_0^1 f(t)\,\mathrm{d}t = 0\},\\
U &\coloneqq \{f \in P(\mathbb{R}) \mid \sin(f(2)) = 0\},\\
V &\coloneqq \{f \in P(\mathbb{R}) \mid f(2) = 0\}.
\end{aligned}
$$

(12)

(b) (i) Let G be a group and let $g \in G$. Give the definition of the *order* of G.

(ii) Prove or disprove the following. For every group G, if every element of G has finite order, then G has finite order.

(iii) Let G be a group. Prove that if every non-identity element of G has order 2, then G is abelian. (13)

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